ONSPOT HACKATHON · INDIC PROMPT COMPRESSOR

Same answer. A fraction of the tokens.

A middleware layer that shrinks Hindi, Kannada and Tamil prompts before they reach the LLM, and checks that the answers stay as good.



58%
₹ saved on DeepSeek V4.1

73%
projected on GLM-5.3

≥ 95%
answer quality kept

Hindi · Kannada · Tamil

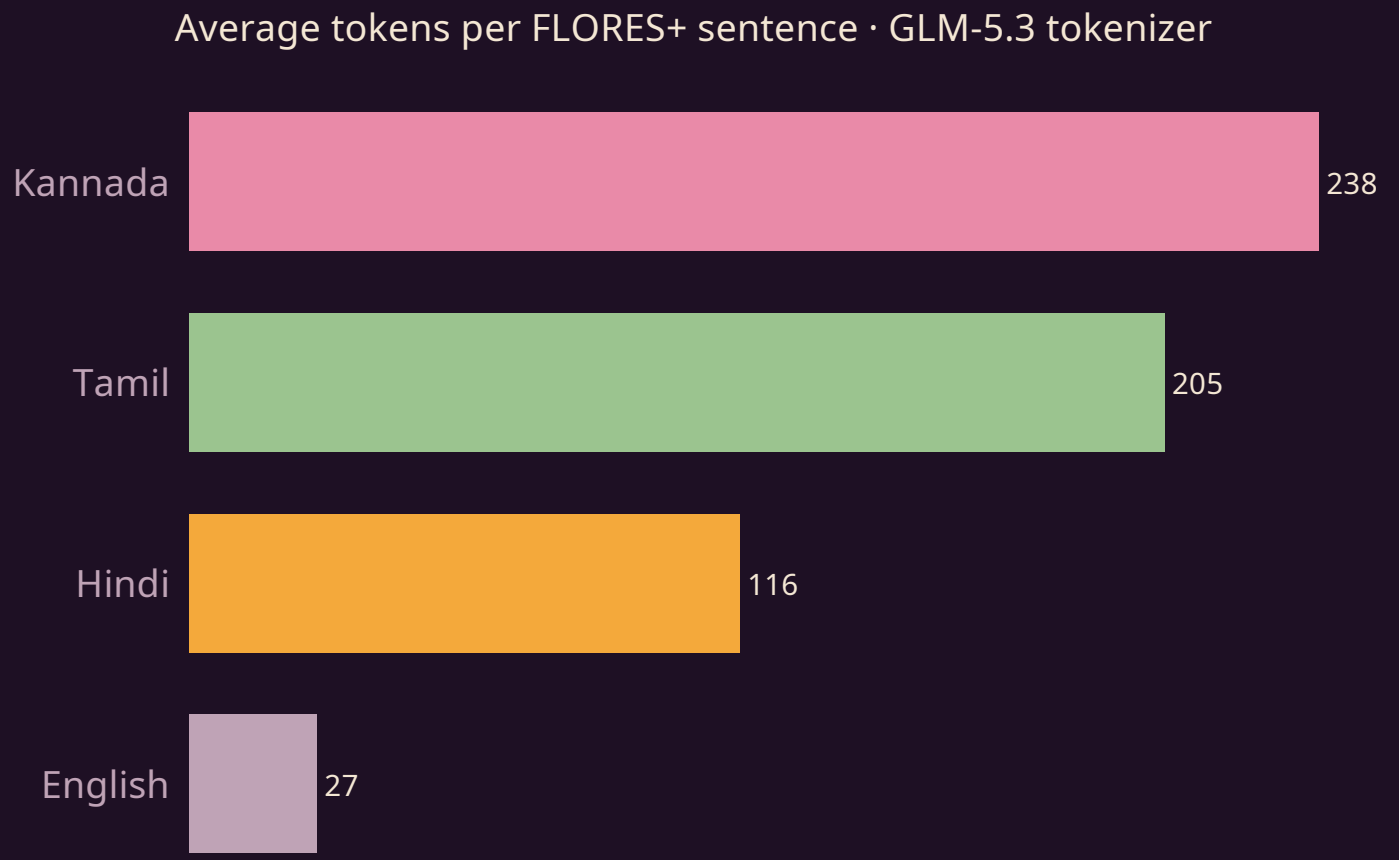
THE PROBLEM

Indic text pays a heavy token tax

8.9×

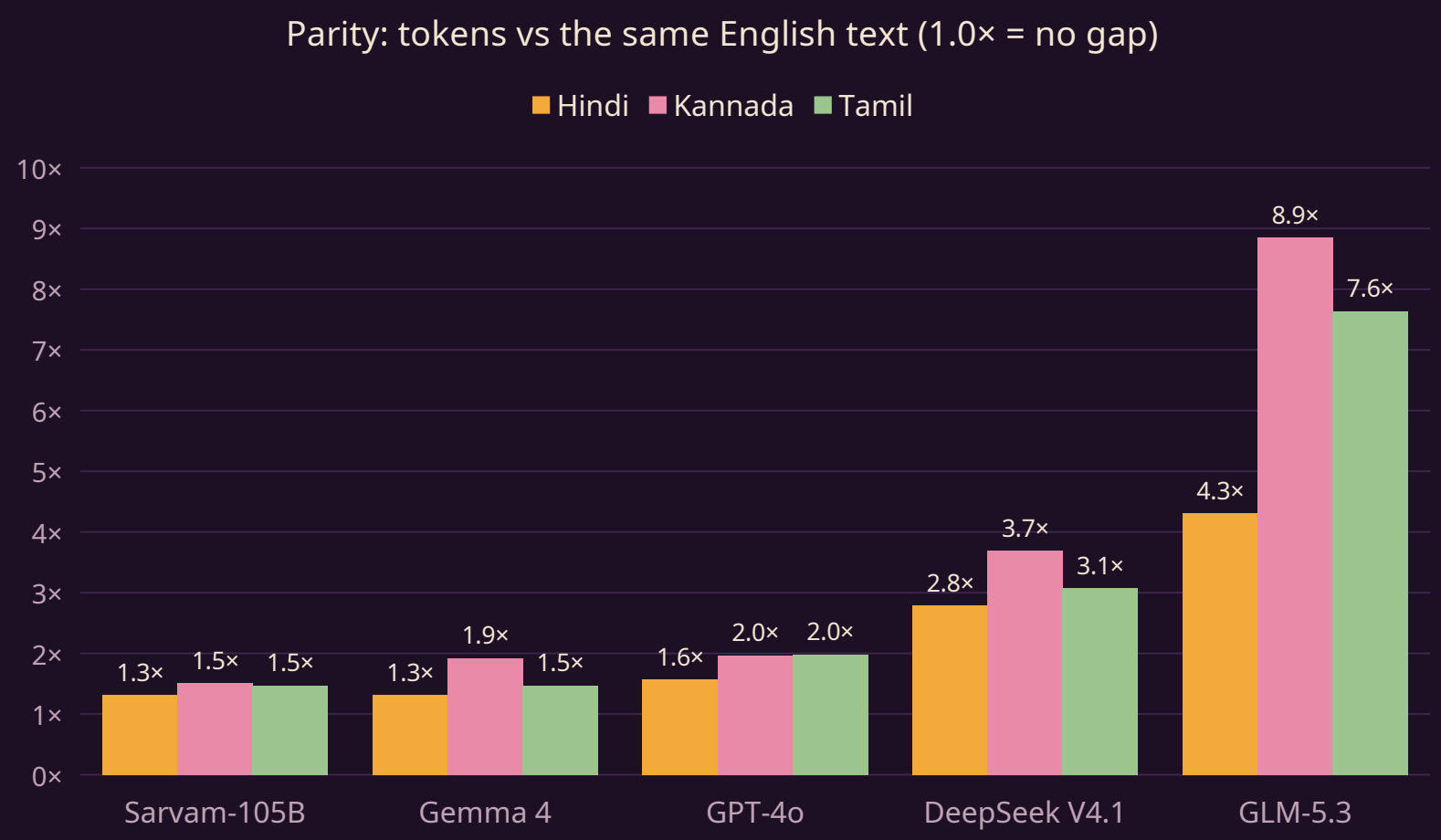
more tokens for the same sentence in Kannada than in English, on the GLM-5.3 tokenizer

Tokens are what you pay for. A Kannada user pays roughly nine English prices for the same question, and the answer comes back just as expensive.



MEASURED GAP · FLORES+ DEVTEST, ~1,000 PARALLEL SENTENCES

The gap depends on the model's tokenizer



Big opportunity

GLM-5.3 (4.3–8.9×) and DeepSeek V4.1 (2.8–3.7×) tokenise Indic scripts badly. That is where compression pays.

Little to gain

Sarvam-105B and Gemma 4 are already close to English (1.3–1.9×). The router only cleans prompts for these models.

01

What we built and tested

A pipeline, twelve compression methods, and a judged experiment of 3,600 answers

THE LAYER

Seven steps between the user and the LLM



01

Detect

Script → Hindi, Kannada or Tamil



02

Clean

Unicode, invisible characters, citations, repeats



03

Route

Rule table + safety guards pick a method



04

Compress

Translate to English and/or prune tokens, locally



05

Call LLM

Sarvam API: DeepSeek, GLM, Sarvam, Gemma



06

Translate

English answer → user's language, locally



07

Report

Tokens, ₹ saved, expected quality

Translation (IndicTrans2), pruning (LLMLingua-2) and sentence matching (LaBSE) run on a 4 GB laptop GPU, so they cost ₹0 and add about 1–4 s per request.

THE MENU

Ten ways to shrink a prompt

M0

Baseline

Original prompt, unchanged. Everything is compared to this

M1

Clean-up

Unicode, zero-width chars, citation marks, spaces, repeated sentences

M2

Prune native

LLMLingua-2 deletes low-value tokens; keep 80 / 60 / 40%

M3

Via English

Translate to English, answer in English, translate back

M3_b5

Careful English

M3 with beam-5 translation: fewer errors, 2× slower

M3h

Hybrid (QA)

English context + 2 key original sentences; answer in original

M4

Romanise

Same language in Latin letters (ITRANS)

M5

English + 60%

Translate, then keep 60% of the English tokens

M5_80

English + 80%

Translate, then keep 80% of the English tokens

M6

English answer

Native input; answer written in English, translated back

M0_rep (the same request sent twice) measures the noise floor. M2 counts as three methods (keep 80 / 60 / 40%), so twelve variants were run in total.

EXPERIMENT DESIGN

How we measured savings and quality

Data

IndicQA: 50 QA + 50 summarisation items per language (frozen, seed 42). FLORES+ for the gap

Target model

DeepSeek V4.1 Flash via Sarvam API, thinking off, temperature 0

Judge

sarvam-105b, blind to the method, always reads the original text. QA: correct / partial / wrong. Summary: faithfulness + coverage + fluency

Pass rule

Quality retained = judge mean vs original ≥ 95% (τ). Strict: 95% bootstrap CI lower bound ≥ 95%

3,600

judged answers in the final run

±2 pts

noise when the same prompt is re-sent

₹94

spent on the target model for 3,600 calls

FOUR RUNS

Each run fixed what the last one exposed

Run 1

Smoke · n = 5

The judge thought for 4,000+ tokens and returned nothing for 26% of answers.

Fix: judge with thinking off + retry

Run 2

Smoke · n = 5

0 missing verdicts. Via English looked strong; one QA item = 20 points.

Next: scale to 50 items

Run 3

Full · n = 50

M3 passes 5 of 6 cells. M5 cheapest but drops facts. Tamil QA fails (0.90).

Next: fix Tamil, soften M5

Run 4

Full · n = 50

M5_80 is the sweet spot. Beam-5 helps summaries. The hybrid fixes Tamil names.

Result: the rule table

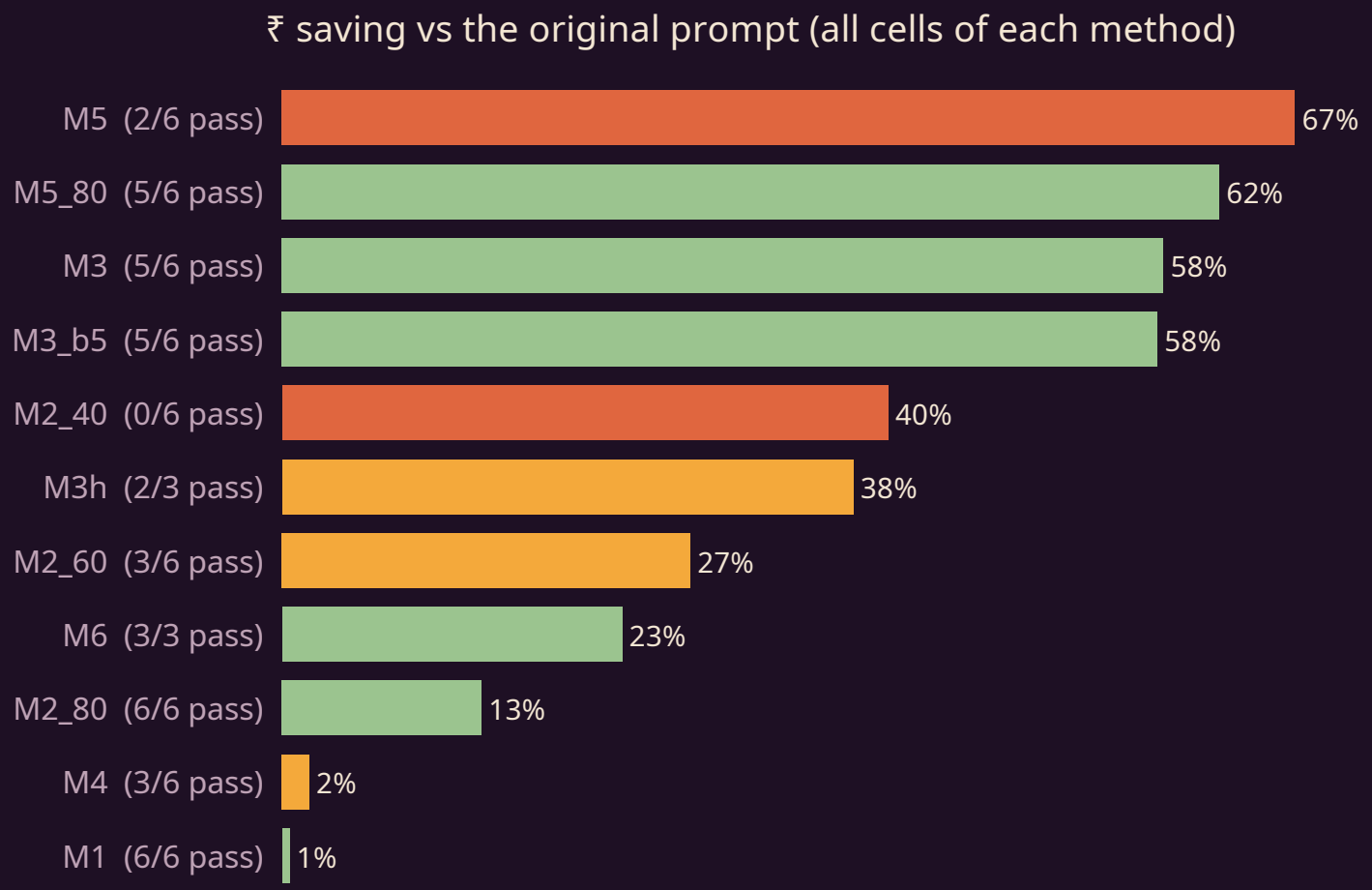
02

Results

DeepSeek V4.1 Flash · 50 items per language × task · τ = 95%

SAVINGS BY METHOD

Going through English saves the most



How to read it

- passes τ in ≥ 80% of its cells
- passes in about half
- fails in most cells

The three English-route methods (M5_80, M3, M3_b5) save 58–62% and keep ≥ 95% quality in 5 of 6 cells.

M5 saves 67% but loses facts in 4 cells. Romanisation (M4) saves almost nothing on DeepSeek.

EVERY METHOD IN EVERY CELL

The quality map: saving / quality kept

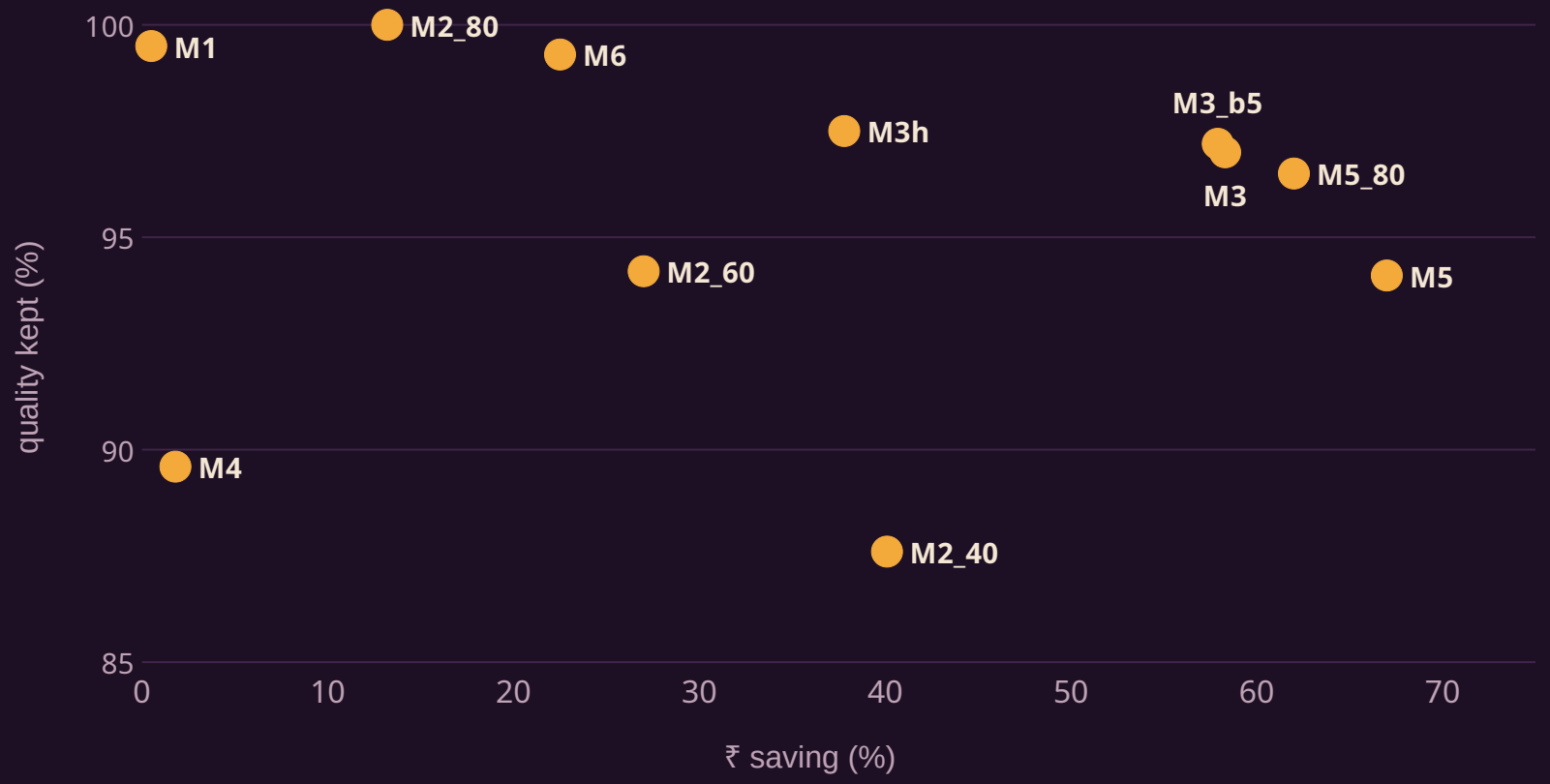
Method	Hindi QA	Hindi summary	Kannada QA	Kannada summary	Tamil QA	Tamil summary
M1	**0% · 1.00**	**0% · 1.00**	**2% · 1.00**	**2% · 1.00**	0% · 0.97	**-1% · 1.01**
M2_80	**16% · 0.99**	11% · 0.97	**17% · 1.03**	**12% · 1.00**	**17% · 1.01**	**11% · 1.00**
M2_60	33% · 0.97	23% · 0.97	33% · 0.90	22% · 0.93	35% · 0.92	26% · 0.97
M2_40	51% · 0.86	34% · 0.91	50% · 0.85	32% · 0.83	53% · 0.85	37% · 0.95
M3	55% · 0.97	54% · 0.97	65% · 0.97	62% · 0.99	60% · 0.93	57% · 1.00
M3_b5	55% · 0.96	**54% · 1.00**	64% · 0.97	**62% · 1.00**	60% · 0.90	**57% · 0.99**
M3h	**38% · 0.99**	–	34% · 0.99	–	41% · 0.95	–
M4	**6% · 1.01**	**3% · 0.98**	24% · 0.99	9% · 0.95	-15% · 0.72	-7% · 0.73
M5	**70% · 1.00**	63% · 0.94	75% · 0.91	67% · 0.92	73% · 0.89	64% · 0.97
M5_80	62% · 0.97	57% · 0.97	69% · 0.97	64% · 0.98	66% · 0.92	60% · 0.99
M6	–	**20% · 1.00**	–	31% · 0.99	–	17% · 0.99

Strict pass: CI lower bound ≥ 95% Pass on the mean (≥ 95%) Below τ

TRADE-OFF

Savings against quality: the frontier

Mean quality kept (%) vs ₹ saving (%), averaged over each method's cells



Read the top-right

The gridline at 95 is the pass bar (τ).

Above the bar (saving · quality)

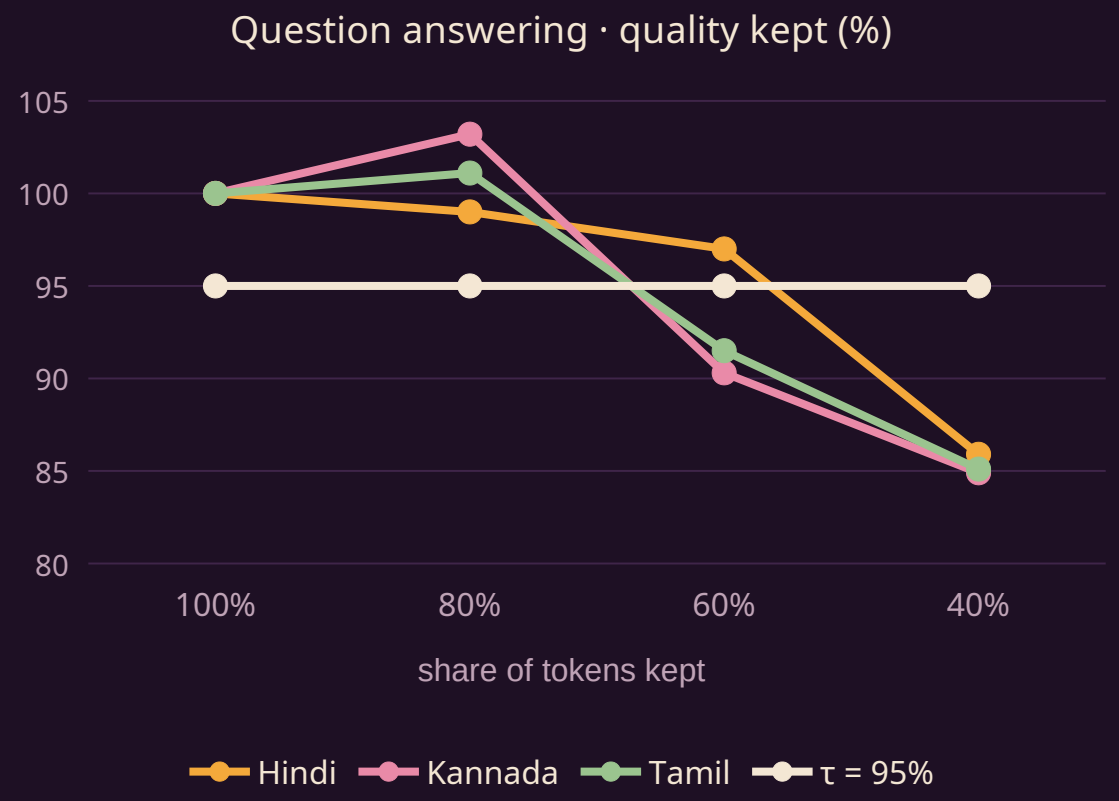
M5_80 62% · 96.5
M3 58% · 97.0
M3_b5 58% · 97.2
M3h 38% · 97.5
M6 23% · 99.3
M2_80 13% · 100.0
M1 1% · 99.5

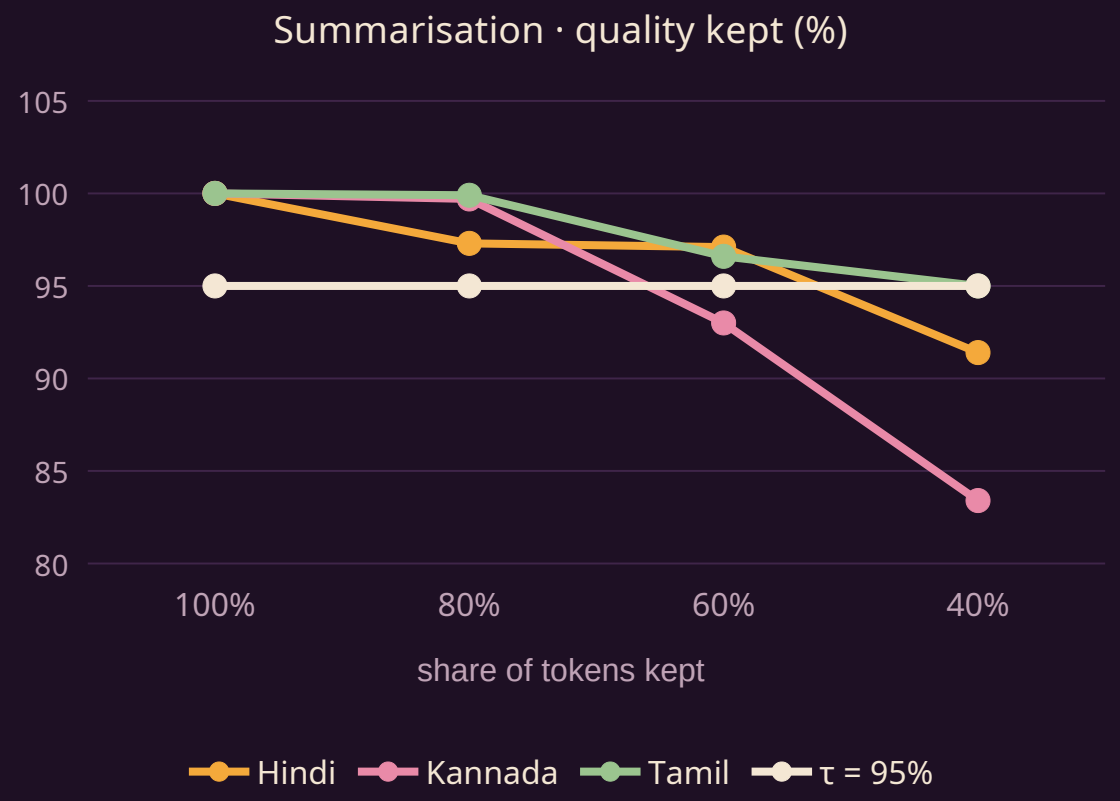
Below the bar

M5 67% · 94.1
M2_60 27% · 94.2
M4 2% · 89.6
M2_40 40% · 87.6

PRUNING IN THE ORIGINAL SCRIPT (M2)

Pruning hurts Kannada and Tamil first





Kannada and Tamil are agglutinative: one long word carries several meanings, so deleting tokens removes meaning. Keeping 80% is safe in every cell, but saves only 11–17%.

THE RULE TABLE

The best method differs by language and task

अ Hindi

Question answering

70% M5

₹ saved · quality 100%

Summarisation

57% M5_80

₹ saved · quality 97%

ಅ Kannada

Question answering

69% M5_80

₹ saved · quality 97%

Summarisation

64% M5_80

₹ saved · quality 98%

அ Tamil

Question answering

17% M2_80

₹ saved · quality 101%

Summarisation

64% M5

₹ saved · quality 97%

58%

₹ saved overall on DeepSeek

73%

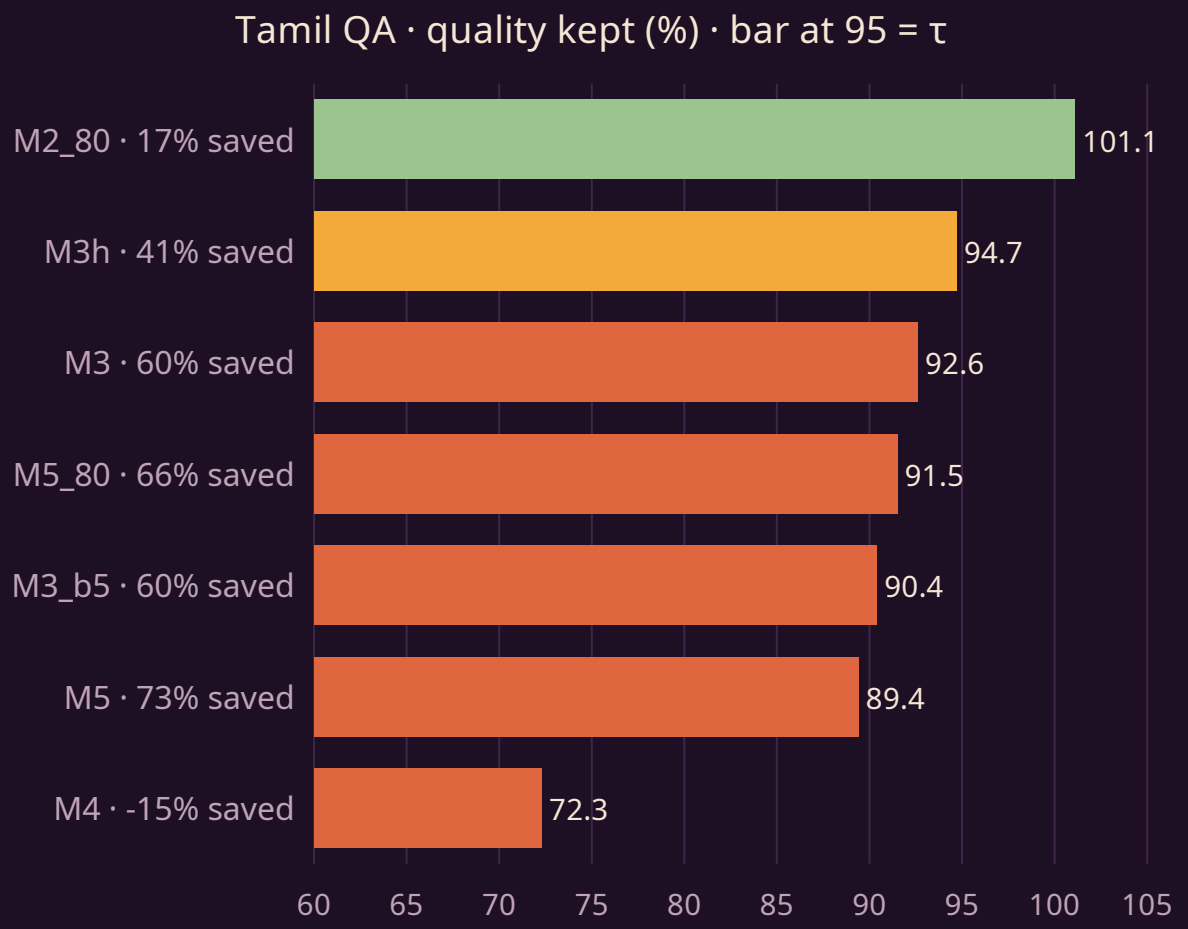
projected on GLM-5.3

51%

under the strict rule (CI ≥ 95%)

THE HARD CASE

Tamil QA: where translation breaks



புற்றுநோய் (cancer)

→ **"Psoriasis"** Translator garbled the key word

ஆர். கே. நாராயணன் (R. K. Narayan)

→ **"K. Narayanan"** Name lost its initials

(answer in the text)

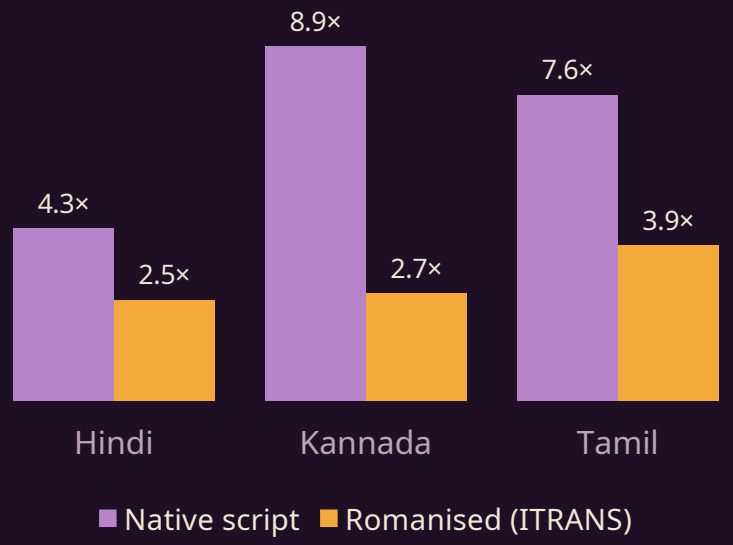
→ **"Not mentioned"** Fact dropped in translation

Fixes tried: beam-5 translation moved the errors around (0.90). The hybrid M3h fixed names but missed other answers (0.947 at 41% saving). The rule table keeps safe pruning (M2_80, 17%).

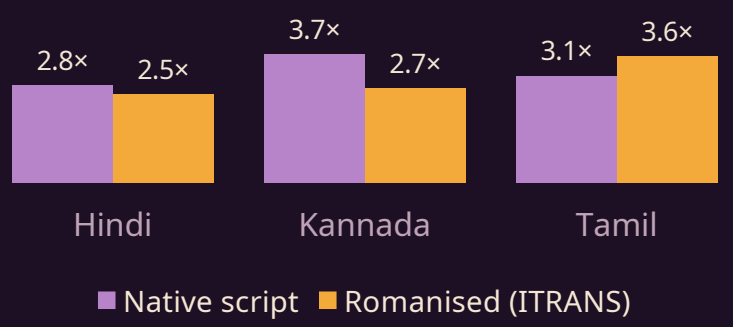
NEGATIVE RESULTS, REPORTED HONESTLY

What did not work

GLM-5.3 parity: romanising helps



DeepSeek parity: barely, and Tamil gets worse



Romanising only helps tokenizers that are bad at Indic scripts (GLM). On Indic-aware tokenizers it adds tokens. Tamil script is ambiguous in Latin letters (ಹ = ka / ga / ha), so the model misreads it.

Romanisation (M4)

Saves 2% on DeepSeek; Tamil costs 15% more and quality falls to 72%

Heavy pruning (M2_40)

Fails every cell: 83–94% quality

Pruning English (M5)

Dropped numbers: “52.52%” became “32%”

English QA answers (M6)

Back-translating 2-word answers broke them; kept for summaries only

03

From experiment to product

A rule-based router and a small web app anyone can try

ROUTING

Rules, not a model, pick the method

1 · Look up the table

Key: target model × language × task × profile. Every DeepSeek row was measured in Run 4. GLM rows reuse it (marked projected). Sarvam-105B and Gemma 4 only get clean-up.

Why rules beat a learned router here

- The per-cell table already gets 58% at τ
- A per-item router could add ~10 points, only for QA
- Laya's checkpoints are English-only and uncalibrated; rules are free, instant and explainable

2 · Apply the guards

- **Code or mostly Latin text** → clean only, never translate
- **Under 120 tokens** → clean only: saving not worth the latency
- **Many digits** → drop the pruning step (M5 → M3)
- **Saves < 10% on this prompt** → fall back to clean-up
- **Answer says “not mentioned”** → re-send the original prompt
- **User picks a method** → override wins

The app: try any prompt, see the saving

Web page
type or load an IndicQA example

FastAPI
/api/preview (free) · /api/run

Local GPU models
IndicTrans2 · LLMLingua-2 · LaBSE, VRAM-budgeted

Sarvam API
DeepSeek · GLM · Sarvam · Gemma + judge

Task · Model · Language

Profile · Method

Context

Question

Preview | Send

M5_80 measured

Rule table: Kannada QA on DeepSeek → M5_80 · 69% ₹ saving, quality 96.8%

327→111	−66%	96.8%
input tokens	₹ cost	expected quality

Answer (translated back) · Compare with original · Judge both

ಗ್ರಾನೈಟ್ ಶಿಲೆಯಿಂದ

Session total: ₹ spent · ₹ saved · tokens saved

LIMITS WE KNOW ABOUT

What these numbers do and don't prove

Temperature 0 is not deterministic

46% of identical DeepSeek prompts got differently worded answers. Differences under ~3 points are noise

The judge is an LLM

Checked against gold-answer F1 and a resend test. The human Hindi spot-check is still pending; no Kannada or Tamil labellers

Wikipedia-style text only

IndicQA passages; real user prompts (chat, code-mixed) were not part of the experiment

n = 50 per cell

CIs are ±5–9 points; per-item summary scores are too noisy to train on

Local latency

Translation adds 1–4 s per request on a 4 GB laptop GPU

GLM is projected

Its saving uses the GLM tokenizer offline; answer quality on GLM is not measured

NEXT

Where this goes next

01	**Validate the judge**	Human Hindi spot-check, deliberately broken answers, and a second judge (Gemma 4)
02	**Measure on more models**	Run the frozen method menu on GLM and Indic-efficient models to replace projections
03	**Fix Tamil QA**	Better sentence selection for the hybrid (top-3, aligned via English) to pass τ at ~40% saving
04	**Real prompts**	Chat and code-mixed prompts; a per-item router for QA if it clearly beats the rules

IN ONE SLIDE

The cheapest prompt is the one that still gets the right answer.



8.9×

the Kannada token tax on GLM-5.3

58%

₹ saved on DeepSeek at ≥ 95% quality

73%

projected saving on GLM-5.3

Thank you